



Gold

△ **Crystalline gold**, made up of octahedral gold crystals

Even before gold became the trading medium of the commercial world and the foundation for modern money, it was prized for its beauty and spiritual significance. To the ancient Egyptians, gold was the perfect material – it offered a glittering yellow surface soft enough to work with, yet it was so durable that it would essentially last forever. In fact, of the three known metals stable enough to have been used for trade at that time, gold was the most suitable. It did not corrode and did not react with other substances. Unlike silver, it did not tarnish and, unlike copper with its high melting point, it could feasibly be melted into currency. Thus it became the most desired metal in the world, transcending geographical borders to become a universal symbol of power, both political and spiritual.

The colour of gold

In its pure state, gold is always golden yellow, but it is too soft to make into jewellery: to increase its hardness, gold is alloyed with other metals. Adding silver, platinum, nickel, or zinc creates pale or white gold. Copper yields red or pink gold, and iron gives a blue tinge. The purity of alloyed gold is expressed in karats, which measures parts per 24: for example, 18-karat gold means there are 18 parts gold out of 24 in the alloy, while 24-karat gold is pure (and mostly too soft to be worn). Note that “karat” is distinct from “carat”, a measurement of weight in gemstones.

Key pieces



Ancient gold | Because of gold's chemically inert nature, ancient gold artefacts buried for thousands of years come out of the ground looking as bright as the day they were made. This Mycenaean brooch is from c.1600–1100 BCE.



Roman gold | From the city of Pompeii buried by volcanic ash and lava in the 1st century CE, this gold armlet is in the form of a coiled snake and shows perfect preservation. The detail on the head and scales on the skin indicate high-quality craftsmanship.

Specification

Chemical name Gold | **Formula** Au | **Colours** Gold yellow

Structure Isometric | **Hardness** 2.5–3 | **SG** 19.3

RI n/a | **Lustre** metallic



Locations

1 Canada **2** USA **3** Brazil **4** South Africa **5** Russia
6 China **7** Australia (and many more)



Cartier panther ring | This spectacular open-form, unisex ring in 18-karat yellow gold from the *Panthère de Cartier* collection features peridot eyes, an onyx nose, and lacquer accents. The wearer's finger goes through the open mouth.